AO detector-level extension parameter-source inventory

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Scope: public caches, derived calculations, synthetic model parameters, derived result artifacts, runtime records, and sources explicitly not claimed as used. Direct public data are separated from synthetic AO proxies.

1. Direct public data caches used

item	file	fields	exact values	source class	source	identifier
SVO 2MASS J filter	data/public/svo_2mass_j_direct.csv	wavelength_m, transmission	107 samples; wavelength range 1.062e-06-1.450e-06 m; effective wavelength 1.241052 micron	direct_public_data	SVO Filter Profile Service direct download for 2MASS/2MASS.J; profile reference http://www.ipac.caltech.edu/2mass/releases/allsky/doc/sec6_4a.html#rsr; 2MASS canonical paper DOI 10.1086/498708.	https://svo2.cab.inta-csic.es/theory/fps/fps.php?ID=2MASS%2F2MASS.J
SVO 2MASS H filter	data/public/svo_2mass_h_direct.csv	wavelength_m, transmission	58 samples; wavelength range 1.289e-06-1.914e-06 m; effective wavelength 1.651366 micron	direct_public_data	SVO Filter Profile Service direct download for 2MASS/2MASS.H; profile reference http://www.ipac.caltech.edu/2mass/releases/allsky/doc/sec6_4a.html#rsr; 2MASS canonical paper DOI 10.1086/498708.	https://svo2.cab.inta-csic.es/theory/fps/fps.php?ID=2MASS%2F2MASS.H
SVO 2MASS Ks filter	data/public/svo_2mass_ks_direct.csv	wavelength_m, transmission	76 samples; wavelength range 1.900e-06-2.399e-06 m; effective wavelength 2.165631 micron	direct_public_data	SVO Filter Profile Service direct download for 2MASS/2MASS.Ks; profile reference http://www.ipac.caltech.edu/2mass/releases/allsky/doc/sec6_4a.html#rsr; 2MASS canonical paper DOI 10.1086/498708.	https://svo2.cab.inta-csic.es/theory/fps/fps.php?ID=2MASS%2F2MASS.Ks
IRSA 2MASS PSC photometry	data/public/target_photometry_2mass_psc_demo_ngs_bright.csv	target_id, ra_deg, dec_deg, J/H/Ks, distance, ph_qual	6 rows; nearest 2MASS_05343359-0523099 J/H/Ks=11.282/10.782/10.719 mag; ph_qual=AAA	direct_public_data	IRSA 2MASS All-Sky Point Source Catalog PSC fp_psc cone query; 2MASS canonical paper DOI 10.1086/498708.	https://irsa.ipac.caltech.edu/cgi-bin/Gator/nph-query?catalog=fp_psc&spatial=cone&radius=60&radunits=arcsec&objstr=83.6331+-5.3911&outfmt=1
MAST Pan-STARRS DR2 photometry	data/public/target_photometry_panstarrs_dr2_demo_ngs_bright.csv	target_id, ra_deg, dec_deg, nDetections, g/r/i/z/y	19 usable rows; photon anchor PS1_101500836297539800 m700=18.871053 mag; photons=0.010434989 per subap/frame	direct_public_data	MAST Pan-STARRS DR2 mean catalog cone query; Pan-STARRS1 Surveys arXiv:1612.05560 and database/data-products arXiv:1612.05243.	https://catalogs.mast.stsci.edu/api/v0.1/panstarrs/dr2/mean.csv?ra=83.6331&dec=-5.3911&radius=0.02&nDetections.gte=1&pagesize=20&columns=%5BobjID%2CobjName%2CraMean%2CdecMean%2CnDetections%2CgMeanPSFMag%2CrMeanPSFMag%2CiMeanPSFMag%2CzMeanPSFMag%2CyMeanPSFMag%2CgMeanPSFMagErr%2CrMeanPSFMagErr%2CiMeanPSFMagErr%2CzMeanPSFMagErr%2CyMeanPSFMagErr%5D
ESO Paranal ASM nighttime snapshot	data/public/eso_asm_paranal_20240729_0300_0800_snapshot.json	seeing, r0_500, tau0, theta0, turbulence speed	UTC 2024-07-29T03:00:00Z to 2024-07-29T08:00:00Z; approximately 23:00-04:00 CLT for Chile winter; seeing=0.7235 arcsec; r0_500=0.139695715 m; tau0=0.003409 s; theta0=1.894 arcsec; turbulence_speed=10.84 m/s; samples=184	direct_public_data	ESO Paranal ASM API direct JSON query over 2024-07-29T03:00:00Z to 2024-07-29T08:00:00Z; fields dimm_paranal-fwhm,mass_paranal-tau0,mass_paranal-tet0,mass_paranal-turb_speed. r0_500_m is derived from median seeing using Fried DOI 10.1364/JOSA.56.001372.	https://www.eso.org/asm/api/?from=2024-07-29T03%3A00%3A00Z&to=2024-07-29T08%3A00%3A00Z&fields=dimm_paranal-fwhm%2Cmass_paranal-tau0%2Cmass_paranal-tet0%2Cmass_paranal-turb_speed
ESO Paranal ASM nighttime time series	data/public/eso_asm_paranal_20240729_0300_0800_timeseries.csv	unix_time_ms, seeing, tau0, theta0, turbulence speed	161 samples over 2024-07-29T03:00:00Z to 2024-07-29T08:00:00Z	direct_public_data	ESO Paranal ASM API direct JSON query over 2024-07-29T03:00:00Z to 2024-07-29T08:00:00Z; fields dimm_paranal-fwhm,mass_paranal-tau0,mass_paranal-tet0,mass_paranal-turb_speed.	https://www.eso.org/asm/api/?from=2024-07-29T03%3A00%3A00Z&to=2024-07-29T08%3A00%3A00Z&fields=dimm_paranal-fwhm%2Cmass_paranal-tau0%2Cmass_paranal-tet0%2Cmass_paranal-turb_speed

2. Derived calculations from public data

calculation	formula or assumption	input	output	source
Fried r0 conversion	r0_500_m = 0.98 * 500e-9 / seeing_rad	ESO ASM median DIMM seeing at 500 nm	0.139695715 m	Fried parameter; DOI 10.1364/JOSA.56.001372
SVO J/H/Ks effective wavelengths	lambda_eff = integral(lambda*T dlambda) / integral(T dlambda)	SVO 2MASS/2MASS.J, H, Ks transmission samples	J=1.241052 micron; H=1.651366 micron; Ks=2.165631 micron	SVO FPS direct caches; 2MASS canonical paper DOI 10.1086/498708
Pan-STARRS 700 nm WFS photon estimate	AB fnu=3631 Jy*10^(-0.4*m); bandwidth=150 nm; D=2 m; throughput=0.25; exposure=1 ms; n_subap=25	PS1_101500836297539800 m700=18.871053	0.010434989 photons/subap/frame	MAST Pan-STARRS DR2 cache; AB magnitude convention from Oke & Gunn DOI 10.1086/160817; engineering estimate, not WFS telemetry
Public-data-informed phase amplitude	phase_amplitude_nm = 260 nm * seeing_arcsec / 0.80 arcsec	ESO ASM seeing=0.7235 arcsec	235.1375 nm for paranal_night_asm	Synthetic scaling anchored to ESO ASM; not a measured wavefront sequence

calculation	formula or assumption	input	output	source
Science Strehl metric	Marechal proxy Strehl = exp(-(2*pi*OPD_rms/lambda)^2)	Synthetic residual OPD RMS and direct SVO J/H/Ks effective wavelengths	J/H/Ks Strehl columns in the science, error-budget, fast-integration, and public-data-informed result CSVs	Analytical AO diagnostic formula; no on-sky PSF calibration is used

3. Notebook 11 observing/error conditions

condition	atmosphere	seeing r0	photons noise	latency	stroke ncpa	misregistration	source
nominal_synthetic	synthetic nominal seeing proxy	0.8 arcsec; r0=0.126337 m	8000 photons/subap/frame; read=1 e-	1 frames; total=0 s	stroke=1000 nm; NCPA=15 nm	shift=(0, 0) px; rot=0 deg; mag=1; shear=0	Notebook 11 nominal synthetic condition for regression comparison.
paranal_night_asm	ESO ASM nighttime direct_public_data cache	0.7235 arcsec; r0=0.139696 m	200 photons/subap/frame; read=1 e-	1 frames; total=0.001 s	stroke=500 nm; NCPA=25 nm	shift=(0, 0) px; rot=0 deg; mag=1; shear=0	Notebook 11 public-data-informed synthetic AO condition: atmosphere and/or catalog photometry may come from public caches, while DM, detector, latency, NCPA, registration, and RTC behaviour remain synthetic engineering proxies.
poor_seeing	ESO ASM nighttime scaled engineering poor-seeing proxy	1.15 arcsec; r0=0.087887 m	200 photons/subap/frame; read=1.5 e-	2 frames; total=0.002 s	stroke=220 nm; NCPA=35 nm	shift=(0.4, 0) px; rot=0.15 deg; mag=1; shear=0	Notebook 11 public-data-informed synthetic AO condition: atmosphere and/or catalog photometry may come from public caches, while DM, detector, latency, NCPA, registration, and RTC behaviour remain synthetic engineering proxies.
faint_ngs	ESO ASM nighttime direct_public_data cache	0.7235 arcsec; r0=0.139696 m	0.010435 photons/subap/frame; read=3 e-	2 frames; total=0.002 s	stroke=180 nm; NCPA=45 nm	shift=(0.5, 0) px; rot=0.2 deg; mag=1; shear=0	Notebook 11 public-data-informed synthetic AO condition: atmosphere and/or catalog photometry may come from public caches, while DM, detector, latency, NCPA, registration, and RTC behaviour remain synthetic engineering proxies.
stress_all_effects	ESO ASM nighttime scaled engineering stress proxy	1.25 arcsec; r0=0.080856 m	0.010435 photons/subap/frame; read=5 e-	3 frames; total=0.003 s	stroke=120 nm; NCPA=70 nm	shift=(0.8, 0.3) px; rot=0.3 deg; mag=1.01; shear=0.005	Notebook 11 public-data-informed synthetic AO condition: atmosphere and/or catalog photometry may come from public caches, while DM, detector, latency, NCPA, registration, and RTC behaviour remain synthetic engineering proxies.

4. Synthetic and mixed model parameters actually used

subsystem	parameters	source class	source
Mode presets	fast: 52 pupil px, 5 lenslets, 5 actuators across, 12 steps, 8000 photons; portfolio: 72 px, 7 lenslets, 7 actuators, 18 steps, 7000 photons; research: 96 px, 9 lenslets, 9 actuators, 30 steps, 6000 photons	synthetic_assumed	IntegrationConfig presets; modes control numerical scale only
Fast detector-level SCAO geometry	D=2.0 m; detector_window_px=18; pad_factor=3; WFS wavelength=700 nm; frame_rate=1000 Hz	synthetic_assumed	IntegrationConfig fast preset and ShwfsGeometryConfig
Detector/WFS noise parameters	Condition matrix controls photons, read_noise_e, latency, stroke, NCPA, and registration stress; DetectorConfig also supports dark current, background, full-well clipping, bad-pixel masks, PRNU, exposure time, and QE	synthetic_assumed plus direct_public_data conditioning	figures/detector_level_SCAO/public_data_informed_conditions.csv; tests/test_detector_centroids.py
DM model	Gaussian influence functions; fast n_actuators_across=5; coupling_width_pitch=0.40; nominal stroke_limit_nm=1000	synthetic_literature_inspired	DM influence-function modelling motivated by arXiv:2306.10803; not measured DM calibration
Interaction matrix / reconstructor	Central-difference poke amplitude=10 nm; rcond scan grid=1e-8,1e-6,1e-4,1e-3; fast-integration kept_modes=13	synthetic_assumed	Self-calibrated detector-level poke matrix; no observatory control matrix is used
compact poke-matrix diagnostic	Command-line diagnostic matrix shape=90x16, rank=16, kept_modes=16, selected rcond=3e-3	synthetic_assumed	figures/detector_level_SCAO/poke_matrix_singular_values.csv; compact detector-level sanity check, not high-order observatory reconstructor conditioning
Science bandpasses	J/H/Ks use SVO 2MASS direct caches when present; top-hat fallback only if a cache is missing	direct_public_data with documented fallback path	data/public/svo_2mass_j_direct.csv, h_direct.csv, ks_direct.csv
fast reference run	open_rms=77.164249 nm; closed_rms=59.442573 nm; H Strehl=0.950455; kept_modes=13; validation=6/6	synthetic_assumed	Fast end-to-end detector-level 2 m SCAO integration using local synthetic/literature-inspired fixtures; not calibrated observatory AO telemetry.

5. Public-data-informed photon scan results

case	photons	closed rms nm	h strehl	command rms nm	saturated frac	provenance
Pan-STARRS direct estimate	0.010435	infeasible	infeasible	infeasible	infeasible	photon=direct_public_data; loop=synthetic_assumed
engineering 50 photons	50	infeasible	infeasible	infeasible	infeasible	photon=synthetic_assumed; loop=synthetic_assumed
engineering 200 photons	200	73.609571	0.924606	82.978653	0.230769	photon=synthetic_assumed; loop=synthetic_assumed
Nominal 8000 photons	8000	56.370882	0.955186	72.878359	0.230769	photon=synthetic_assumed; loop=synthetic_assumed

6. Public-data-informed scenario results

condition	enabled effects	closed rms nm	h strehl	valid centroid frac	command rms nm	saturated frac	decomposition
nominal_synthetic	synthetic_conditioned_phase_sequence+detector_noise+latency_1_frames+science_path_ncpa	18.313289	0.995123	1	113.019523	0	WFS/science no-NCPA proxy=10.506025 nm; plus NCPA=18.313289 nm
paranal_night_asm	eso_asm_conditioned_synthetic_phase_sequence+detector_noise+latency_1_frames+dm_stroke_limit+science_path_ncpa	68.265238	0.934878	1	143.679588	0	WFS/science no-NCPA proxy=63.522773 nm; plus NCPA=68.265238 nm
poor_seeing	eso_asm_conditioned_synthetic_phase_sequence+detector_noise+latency_2_frames+dm_stroke_limit+wfs_dm_misregistration_proxy+science_path_ncpa	86.783779	0.897049	1	133.163689	0.153846	WFS/science no-NCPA proxy=79.412999 nm; plus NCPA=86.783779 nm
faint_ngs	eso_asm_conditioned_synthetic_phase_sequence+detector_noise+latency_2_frames+dm_stroke_limit+wfs_dm_misregistration_proxy+science_path_ncpa	infeasible	infeasible	infeasible	infeasible	infeasible	infeasible
stress_all_effects	eso_asm_conditioned_synthetic_phase_sequence+detector_noise+latency_3_frames+dm_stroke_limit+wfs_dm_misregistration_proxy+science_path_ncpa	infeasible	infeasible	infeasible	infeasible	infeasible	infeasible

7. Long-run runtime records

script	started utc	finished utc	runtime minutes	limit minutes	within limit	source
examples/run_public_data_informed_ao_demo.py	2026-07-20T20:16:15Z	2026-07-20T20:26:46Z	10.51729	30	True	Local wall-clock runtime for the public-data-informed Notebook 11 demo. The run uses tracked public caches and synthetic AO proxies; no live archive query is performed.

8. H-band science metrics

case	opd rms nm	h strehl	source class
open_loop	130.503151	0.780666	synthetic_assumed
ideal_closed_loop	0	1	synthetic_assumed
realistic_closed_loop	28.710693	0.988057	synthetic_assumed

9. Error-budget scenario results

scenario	closed rms nm	h strehl	command rms nm	saturated frac	source class
ideal_static	1.052463	0.999983	117.536223	0	synthetic_assumed
dynamic_multilayer_proxy	7.207515	0.999242	89.600338	0	synthetic_assumed
detector_noise	7.498357	0.999177	89.906955	0	synthetic_assumed
latency	20.375744	0.993964	122.929464	0	synthetic_assumed
stroke_limit	11.97703	0.997881	68.736172	0.230769	synthetic_assumed
misregistration	9.176844	0.998771	89.600338	0	synthetic_assumed
ncpa	45.328955	0.970492	89.600338	0	synthetic_assumed
all_effects	50.805583	0.963137	75.601849	0.230769	synthetic_assumed

Validation scope note: the public-data-informed checks below confirm public-data provenance, finite metrics, cache presence, and runtime; they are not an adaptive-optics performance validation. A faint scenario with valid_centroid_frac = 0 has no usable WFS centroids, so its loop is frozen (closed RMS approaches open-loop) yet still passes these checks.

10. Public-data-informed validation checks

check	passed	metric	tolerance	source class	message
nighttime_eso_asm_condition_present	True	4	1	direct_public_data	At least one public-data-informed condition uses the nighttime ESO ASM cache.
catalog_photon_condition_present	True	2	1	direct_public_data	At least one condition uses catalog-derived Pan-STARRS photon-budget input.
scenario_metrics_finite	True	1	1	synthetic_assumed	Public-data-informed scenario metrics are finite for every calibration-feasible condition.
infeasible_conditions_recorded	True	2	5	synthetic_assumed	Conditions whose photon budget cannot calibrate the WFS are recorded as infeasible with an explicit reason.
internal_ao_terms_not_direct_public	True	1	1	synthetic_assumed	Scenario rows do not claim synthetic AO internals as direct public data.
jhk_svo_direct_caches_expected	True	1	1	direct_public_data	J/H/K science metrics are configured to prefer SVO direct caches.
runtime_under_30m	True	10.51729	30	package_reference	Public-data-informed AO demo runtime stays below the documented 30 minute local-run limit.

11. Generated visual/result artifacts

artifact	contents	basis
figures/detector_level_SCAO/public_data_overview.png	ESO ASM nighttime time series, SVO J/H/Ks filters, catalog field map, optical/NIR photometry anchors	direct public caches
figures/detector_level_SCAO/public_filter_curves_jhk.png	Direct SVO 2MASS J/H/Ks filter curves	direct public SVO caches
figures/detector_level_SCAO/public_data_photon_budget.png	Pan-STARRS 700 nm WFS photon-budget estimate	Pan-STARRS direct data + explicit engineering assumptions

artifact	contents	basis
figures/detector_level_SCAO/public_data_informed_ao_photon_scan.png	AO residual/Strehl/stroke scan conditioned on ESO ASM + Pan-STARRS	direct public conditioning + synthetic loop
figures/detector_level_SCAO/poke_matrix_singular_values.png	Compact detector-level DM/WFS poke-matrix singular spectrum	synthetic detector-level calibration sanity check
figures/detector_level_SCAO/public_data_informed_error_budget.png	Five-condition public-data-informed AO scenario map	direct public conditioning + synthetic AO proxies
figures/detector_level_SCAO/public_data_informed_runtime.csv	Local runtime record for the slower public-data-informed demo	package runtime metadata
figures/detector_level_SCAO/public_data_informed_runtime.json	JSON copy of the public-data-informed demo runtime record	package runtime metadata
figures/detector_level_SCAO/public_data_informed_validation.png	Public-data-informed provenance and metric validation checks	public-data and synthetic-boundary checks
figures/detector_level_SCAO/science_psf_metrics.png	J/H/Ks PSF metrics	SVO J/H/Ks direct caches
figures/detector_level_SCAO/error_budget_scenarios.png	8-row error-budget scenario comparison	synthetic AO scenarios with SVO J/H/Ks science metrics
figures/detector_level_SCAO/fast_error_budget.png	fast all-scenario integration summary	synthetic fast model with SVO J/H/Ks bandpasses
figures/detector_level_SCAO/fast_validation.png	Marechal, diffraction, photon, latency, fitting, reproducibility checks	synthetic validation scans

12. Sources explicitly not claimed as used

source	identifier	status
Gaia Archive / Gaia DR3	https://gea.esac.esa.int/archive/ ; Gaia DR3 DOI 10.1051/0004-6361/202243940	Not used in this run. Archive access failed from this environment; Pan-STARRS DR2 is the optical-photometry substitute.
ERA5 pressure/single levels	DOI 10.24381/cds.bd0915c6 and DOI 10.24381/cds.adbb2d47	Not used. ESO ASM supplies direct seeing/tau0/theta0/turbulence-speed conditioning for this demonstrator; CDS credentials would be required for ERA5.
ESO Science Archive / Keck / Gemini / EIDC images	ESO archive URL, KOA, Gemini API, EIDC arXiv:2101.05080 and arXiv:2410.17636	Not used for current simulated-loop results. No on-sky PSF validation is claimed.
Observatory telemetry, DM influence matrices, RTC logs	No public file in repository	Not used. Detector, DM, latency, NCPA, registration, and control terms are synthetic engineering proxies.

13. Source index

id	source	identifier	use
ESO ASM	ESO Paranal ASM API and ambient query forms	https://www.eso.org/asm/api/	direct public atmosphere cache
SVO FPS	SVO Filter Profile Service	https://svo2.cab.inta-csic.es/theory/fps/	direct public 2MASS J/H/Ks filter curves
2MASS	IRSA 2MASS PSC	https://irsa.ipac.caltech.edu/Missions/2mass.htm l ; DOI 10.1086/498708	direct public J/H/Ks catalog photometry and filter identity
Pan-STARRS	MAST Pan-STARRS DR2 mean catalog	arXiv:1612.05560 and arXiv:1612.05243	direct public optical photometry substitute for Gaia
AB magnitude	AB photon-budget convention	Oke & Gunn DOI 10.1086/160817	700 nm WFS photon estimate from Pan-STARRS magnitudes
Fried r0	Fried parameter conversion	DOI 10.1364/JOSA.56.001372	r0 derived from ESO ASM seeing
DM influence	Gaussian synthetic DM influence functions	arXiv:2306.10803	literature-inspired DM shape choice
Noll	Zernike/statistical aberration background	DOI 10.1364/JOSA.66.000207	background reference for modal aberration vocabulary